COURSE OVERVIEW

# GEOSPATIAL REPRESENTATION LEARNING

Introduction and Kick-off

Winter Semester 2026/27
Mobile Robotics, Geodesy and
Geoinformation & Geodetic Engineering

PRESS → OR SPACE. →

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UNIVERSITÄT BONN

# Three Phases:

### Phase 1: Lectures & Labs

Build conceptual and technical foundations through lectures and hands-on exercises.

**Examination:** final written exam on the course content.

### Phase 2: Paper Analysis

Analyze and review recent research papers in student groups.

**Examination:** pass/fail paper presentation with questions.

### Phase 3: Group Projects

Develop code and conduct a research-oriented geospatial analysis project in groups.

**Examination:** four-page scientific paper.

**Bloom's cognitive levels across the course**

| Remember | Understand | Apply | Analyze | Evaluate | Create |
|---|---|---|---|---|---|

# Timeline at a Glance

**Fridays:** Lectures 10:00–12:00 in HS VIII (Meckenheimer Allee 172)*
Labs 13:00–15:00 in Seminarraum 2.003 (Nußallee 1)

* Lecture 2 on October 30 is in HS IX (same floor).

| Course week | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Event date | Oct 16 | Oct 23 | Oct 30 | Nov 6 | Nov 13 | Nov 20 | Nov 27 | Dec 4 | Dec 11 | Dec 18 | Christmas + Jan 8 | Jan 15 | Jan 22 | Jan 29 | Feb 5 |
| **Phase 1: Lectures** | Intro | L1 | L2* | L3 | L4 | No class | L5 | L6 | | | | | | | |
| **Phase 1: Labs** | | Lab 1 | Lab 2 | Lab 3 | Lab 4 | | | | | | | | | | |
| **Phase 2: Papers** | | | | | | | P1 | P2 | P3 | P4 | | | | | |
| **Phase 3: Projects** | | | | | | | | | | | | Group Projects | | | |

## Presence and Examination Events

**Paper Presentations pass/fail**
Nov 27, Dec 4, 11 & 18, 2026

**Phase 1 Written exam**
February 5th, 2027

**Phase 3 paper deadline**
Date: TBD

# Phase 1: Lectures and Labs

## Lectures

**10:00–12:00 (s.t.)**

HS VIII (Meckenheimer Allee 172)
Lecture 2 on October 30: HS IX (same floor).

Presentation slides introduce the core concepts, terminology, and research context.

They provide the shared conceptual foundation for the hands-on work.

## Labs

**13:00–15:00**

Seminarraum 2.003 (Nußallee 1)

Interactive activities reinforce the lecture through self-directed, collaborative, and hands-on learning:

- Explore representations using PCA and t-SNE
- Collect, clean, and combine geodata
- Train and evaluate machine-learning models
- Investigate model architectures and explain them through peer teaching

# Phase 1: Overview - Lectures & Labs

**Lecture 1**

**Representations**
*What is a representation?*

**Lecture 2**

**Geospatial Data**
*What is geodata?*

**Lecture 3**

**Machine Learning on Data Representations**
*Learning on fixed representations*

**Lecture 4**

**Deep Representation Learning I**
*Learning the representation itself*

**Lecture 5**

**Deep Representation Learning II**
*Learning beyond memorization*

**Lecture 6**

**Geospatial Embeddings**
*Giving location a meaning*

**Lab 1**

**Visualizing high-dimensional representations**

**Lab 2**

**Geodata Scavanger Hunt**

**Lab 3**

**Machine Learning on Geospatial Data.**

**Lab 4**

**Practical: Supervised Deep Learning**

# Phase 2: Paper Analysis

**Paper Analysis slides**
Open the session deck

**PowerPoint presentation template**
Coming soon

## Format

Each topical session introduces a new research area and assigns papers to the student groups.

1. **Read (between classes):** Study the assigned paper with your group.
2. **Analyze (between classes):** Identify the problem, method, evidence, limitations, and impact, and prepare your presentation.
3. **Present (in class):** Present your analysis, answer questions, and discuss the paper with the class.

## Four Topics

| Session | Topic |
|---|---|
| 1 | Implicit Neural Representations and Approximation |
| 2 | Location Encoders |
| 3 | Self-Supervised Representation Learning |
| 4 | Geospatial Embeddings |

# Phase 3: Group Projects

Groups of five · One research question · One four-page scientific paper

### Solar Fields

**Map solar expansion on farmland.**

### Border Landscapes

**Compare farming landscapes across European borders.**

### Plastic Currents

**Explore potential plastic sources and sinks along the Rhine.**

### Earth Economies

**Connect landscape embeddings with production and trade.**

### Embedding Seasons

**Detect double versus triple cropping in southern Brazil.**

### Generative Fields

**Generate satellite imagery from field maps in India.**

**Paper:** Question & hypothesis → Experiment & baselines → Results & limitations
**Reproducibility:** GitHub repository with code and a short README.

Project descriptions and evaluation rubric ↗

# AI policy

## The goal of this course is for you to think.

You are encouraged to use ChatGPT, Claude, Gemini, and other AI tools to:

- Retrieve information and explore new ideas
- Structure **your own thoughts**
- Discuss concepts with an LLM agent to sharpen your understanding

Labs and exercises are designed to be interactive, and the exam is handwritten. AI agents cannot do the thinking or complete these activities for you.

# Who’s in the Room?

# Teachers

**Marc Rußwurm**

Professor

**Sawssen Hannachi**

Doctoral Researcher

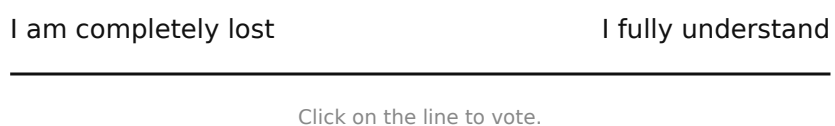
**Gabriele Tijunaityte**

Doctoral Researcher

# What about you? - Live Understanding Pulse

Throughout this course, I will show this widget to touch base on your level of understanding. It synchronizes across all devices, including in-class and virtual participants.

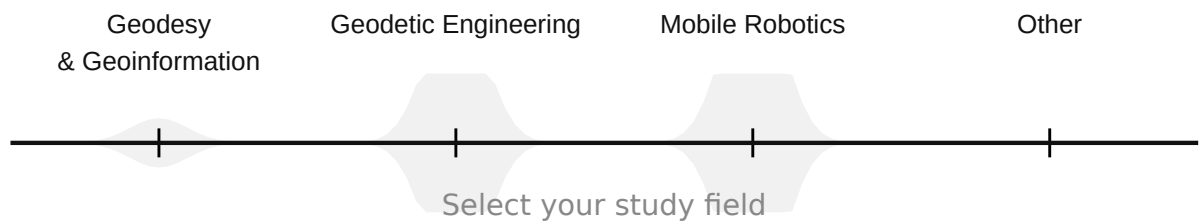
**Try it out - click on the line:**




This gives me feedback during class and also tracks the general understandability of the slides, so I can improve them in future iterations.

# What is your study field?

Click the category that best describes your study programme.



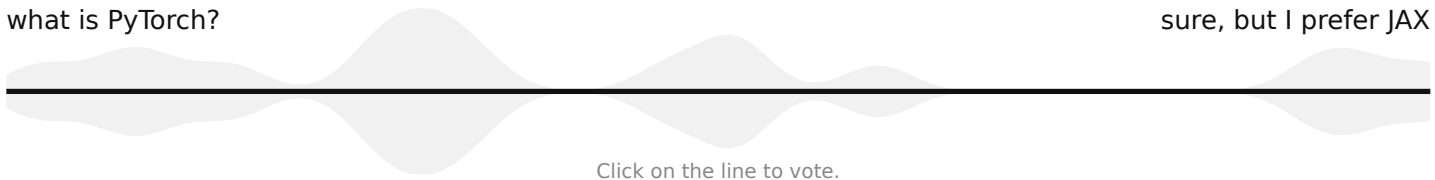

Light gray fill: responses of the 14 participants on 16 October 2026 (10:00–12:00), last vote per person.

# I can train Transformer models

Light gray fill: responses of the 9 participants on 16 October 2026 (10:00–12:00), last vote per person.

# I can write a PyTorch data loader

what is PyTorch? sure, but I prefer JAX



Click on the line to vote.

Light gray fill: responses of the 16 participants on 16 October 2026 (10:00–12:00), last vote per person.

# I can explain coordinate reference systems

coordinate?

sure, which one? WGS84?

Click on the line to vote.

Light gray fill: responses of the 15 participants on 16 October 2026 (10:00–12:00), last vote per person.

# Course Learning Outcomes

**Explain** geospatial representations and embeddings, including their potential and limitations.

**Compare and curate** geospatial datasets from imagery, time series, GIS, and socio-economic sources.

**Implement** Python ML workflows with fixed and learned geospatial representations.

**Evaluate** models for environmental applications: performance, generalization, and task suitability.

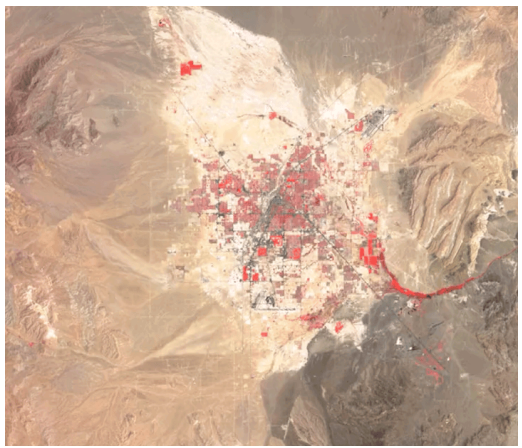
**Critically assess** research on geospatial representations, foundation models, and Earth embeddings.

**Conduct** a group research project and communicate methods, results, and limitations.

Light gray fill: responses of the 16 participants on 16 October 2026 (10:00–12:00), last vote per person.

# Why this course?

# The Earth faces global challenges



Urbanization

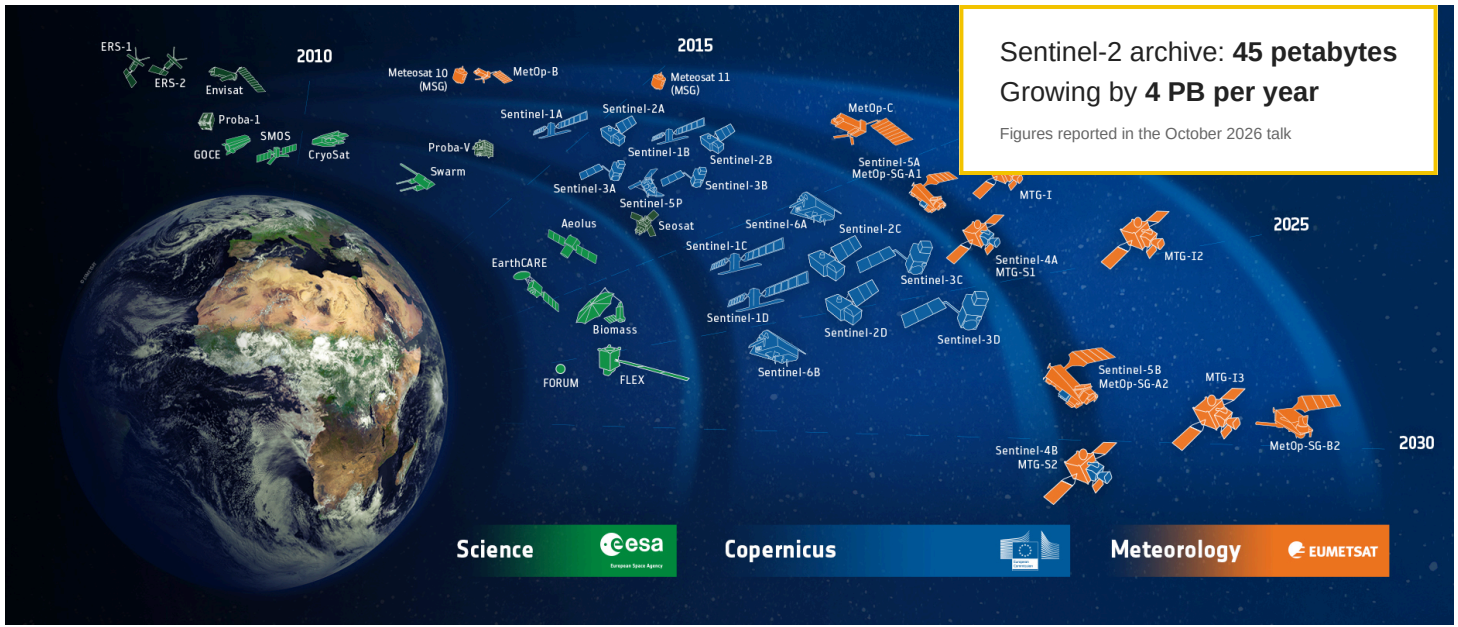
Deforestation

**We need to analyze geodata at large scale**

to inform policies on biodiversity, food security, human health, and livelihoods.

Images: Google Earth Engine Time Series Gallery · Adapted from the Open Earth Monitor talk, October 2026.

# Observing the Earth at unprecedented scales

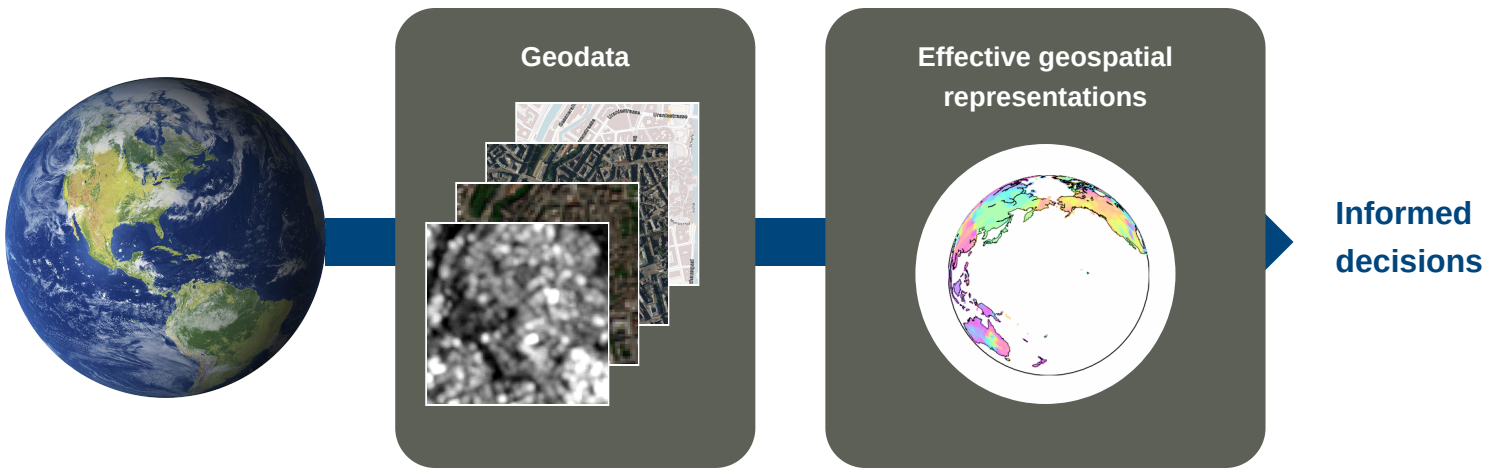



Sentinel-2 archive: **45 petabytes**
Growing by **4 PB per year**

Figures reported in the October 2026 talk

Image: European Space Agency · Archive figures reproduced from the source talk.

# We need effective representations of heterogeneous geodata

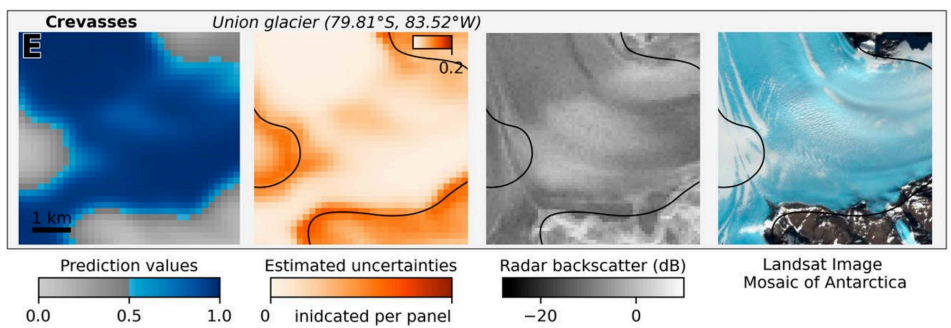


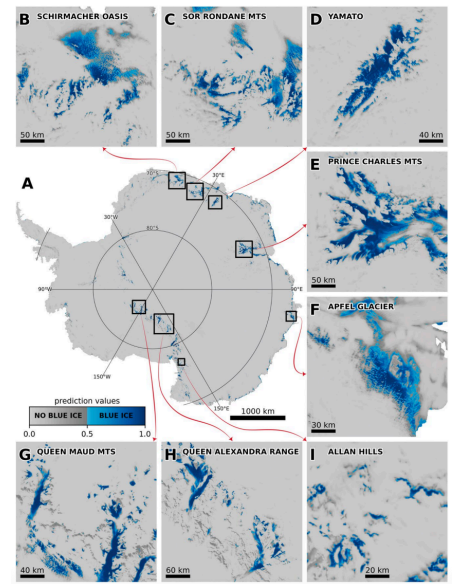

Geospatial data from ground, air, and space at enormous scale

Modern data representations learning with deep neural networks

# Finding meteorites in Antarctic blue ice

Veronica Tollenaar
Vrije Universiteit Brussel




Tollenaar, V., Zekollari, H., Pattyn, F., Rußwurm, M., Kellenberger, B., Lhermitte, S., Izeboud, M., & Tuia, D. (2024). Where the White Continent Is Blue: Deep Learning Locates Bare Ice in Antarctica. *Geophysical Research Letters*, 51(3), e2023GL106285. doi:10.1029/2023GL106285

# Pollution monitoring

Cuttings Beach, Durban, South Africa
Image: Lisa Guastella

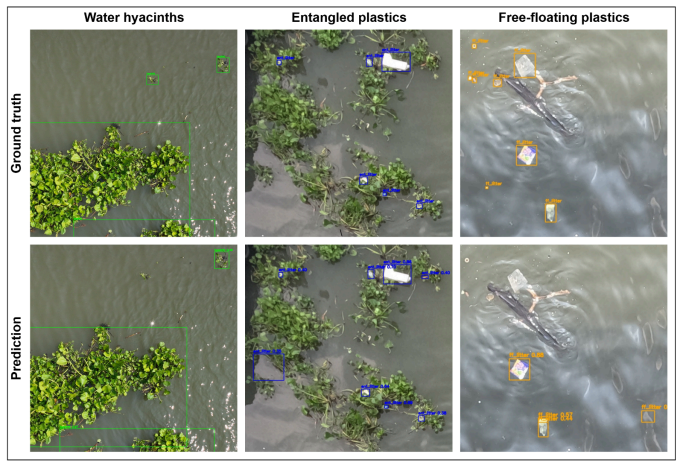



Figure 3: Examples of predicted bounding boxes compared to hand-annotated (ground truth) bounding boxes.

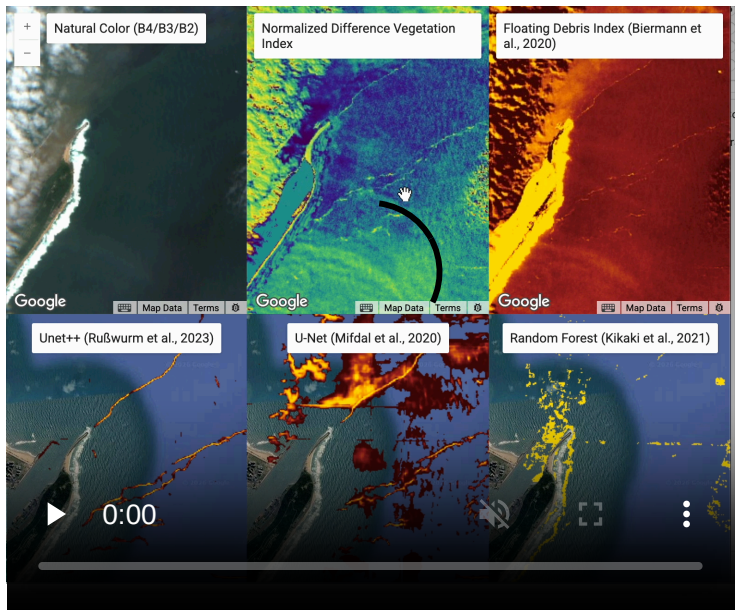
Giel Hagenbeek · Wageningen University & Research

Hagenbeek, G. W., van Emmerik, et al., including Rußwurm, M. (2026). Exploring transferability of plastic-water hyacinth interaction and detection in rivers. *iScience*, 29(6).

# Detecting litter in open waters

Cuttings Beach, Durban · Image: Lisa Guastella

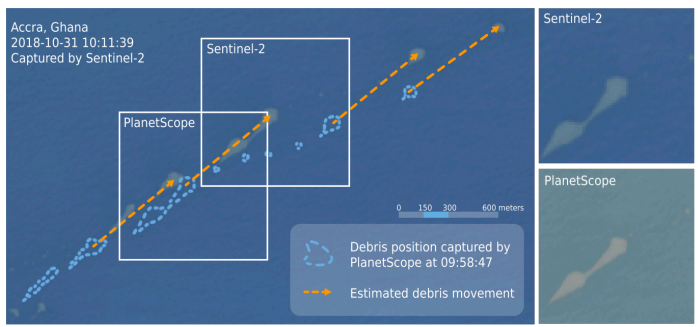
Sentinel-2 satellite

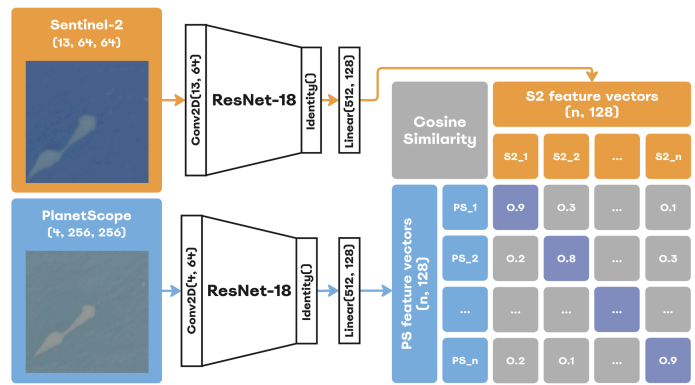



Rußwurm, M., Venkatesa, S. J., & Tuia, D. (2023). Large-scale detection of marine debris in coastal areas with Sentinel-2. *iScience*, 26(12). Marine debris detector

# Tracking litter patches across sensors

Gabrielė Tijūnaitytė · Wageningen, Bonn



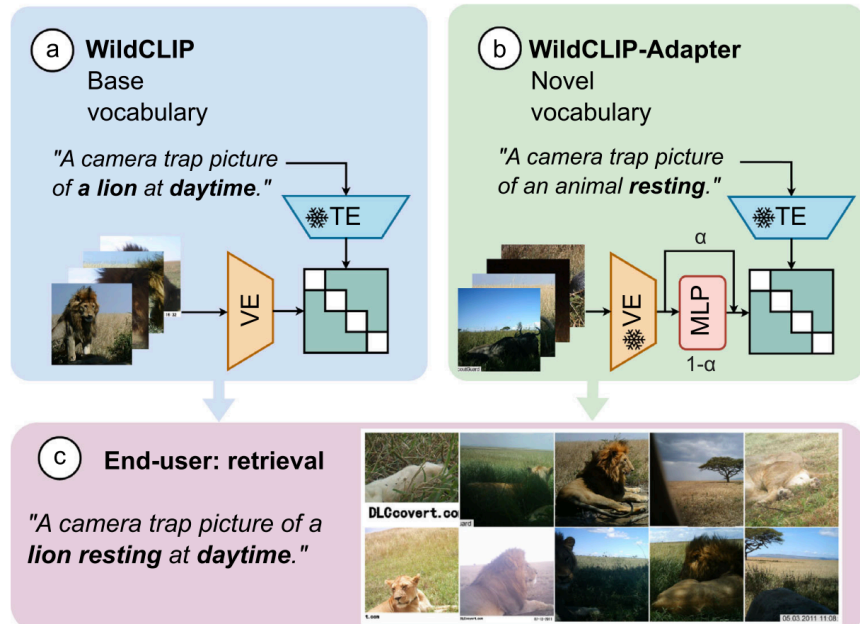




Tijūnaitytė, G., Dalsasso, E., & Rußwurm, M. (2026). Cross-Sensor Matching of Marine Litter Patches in Satellite Double Acquisitions. ECCV Marine Vision Workshop (MaVi).

# Text-based image retrieval on camera traps

Trained on the Snapshot Serengeti camera-trap dataset (Swanson et al., 2015) with more than **seven million images**.

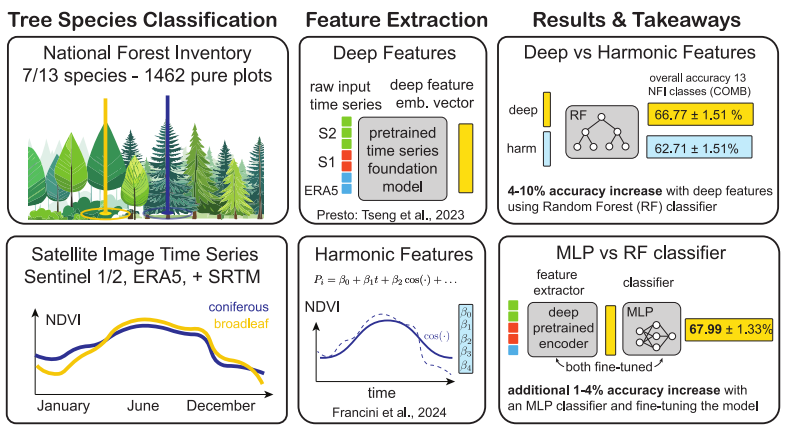
Valentin Gabeff · EPFL

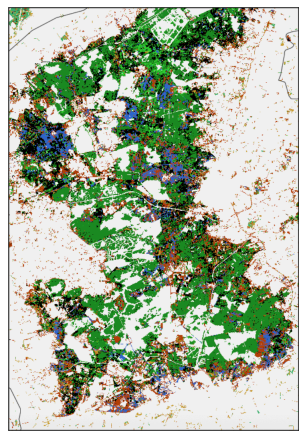


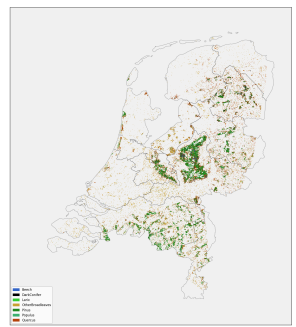

Gabeff, V., Rußwurm, M., Tuia, D., & Mathis, A. (2024). WildCLIP: Scene and animal attribute retrieval from camera trap data with domain-adapted vision-language models. *International Journal of Computer Vision*, 132(9), 3770–3786.

# Tree species classification

Takayuki Ishikawa · WUR MSc student





Ishikawa, T., Bonannella, C., Lerink, B. J. W., & Rußwurm, M. (2026). Assessing the effectiveness of deep embeddings for tree species classification in the Dutch Forest Inventory. *Science of Remote Sensing*, 14, 100498. doi:10.1016/j.srs.2026.100498

# Forest change in the Swiss Alps

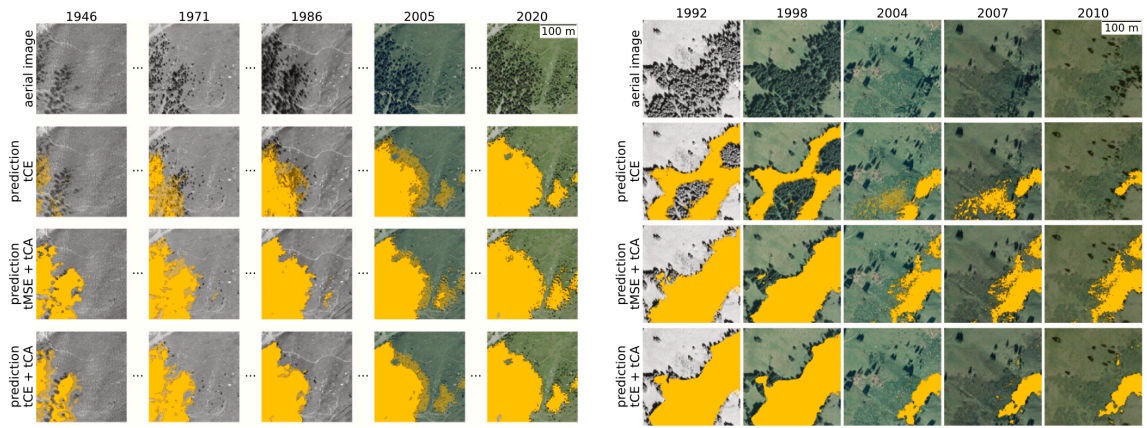
Historic aerial photos (1946–2020) reveal forest gains at high altitudes as the tree line advances upwards with a warming climate.

**Historic aerial imagery**

2,660 tiles

1 km × 1 km

12–20 time steps



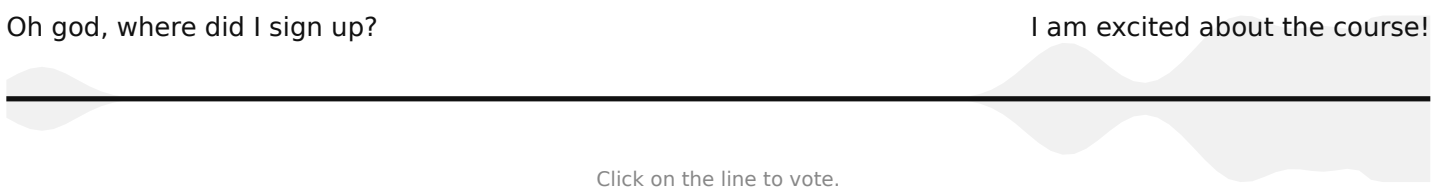

(a) Forest gain (b) Forest loss

# See you next Friday for Lecture 1!

**Representations: What is a representation?**

We will explore how images, words, and places can be represented.

**10:00–12:00 (s.t.) · HS VIII**
Meckenheimer Allee 172

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